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Formal Language & Finite Automata

Undecidability

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All reading is from the "Introduction to the theory of computation" - Micheal Sipser

- Turing machines - Pages 198 - 206 (essential)

Decidable languages

Define the following languages:

$$A_{\text{CFG}} = \{\langle G, w \rangle \mid G \text{ is a CFG and } w \in L(G)\};$$

$$E_{\text{CFG}} = \{\langle G \rangle \mid G \text{ is a CFG and } L(G) = \emptyset\};$$

Theorem

The languages A_{CFG} and E_{CFG} are Turing decidable.

Theorem

Every context-free language A is Turing decidable.

Diagonalization

Definition

A set X is *countable* (or *countably infinite*) if there is a bijection $f : X \rightarrow \mathbb{N}$.

Example

$\mathbb{N}$, $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Theorem

$\mathbb{R}$ is *uncountable*.